S^4D

Self-Taught Semi-Self Speculative Decoding

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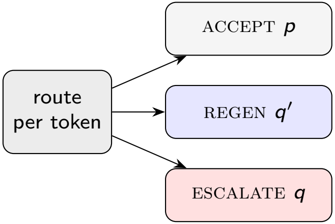
Background and motivation

- **Speculative decoding** accelerates LLM inference: a cheap drafter proposes a block; the target verifies it in one parallel pass. Lossless, bounded by the *acceptance rate*.
- **VIA-SD** adds a **slim intermediate verifier** q' (layer-skipped target), giving three tiers per token — ACCEPT / REGENERATE (q') / ESCALATE (q) — to reduce full-model calls.
- **Limitation:** the tier is chosen by two **hand-tuned thresholds** ($\alpha_1{=}0.5, \alpha_2{=}0.3$): global, static, and sub-optimal ($\leq$ 0.50 even tuned).

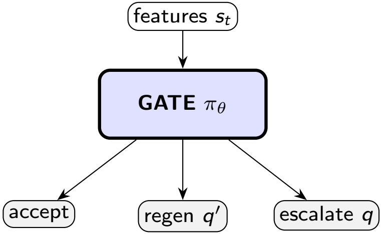


S^4D: learned routing as an MDP

We recast per-token tier selection as a **Markov decision process**:

- **State** s_t: q-free features (drafter/q' confidence, agreement, KL, position).
- **Action** $a_t \in \{\text{accept, regenerate, escalate}\}$.
- **Policy** π_θ: a small MLP.
- **Reward** $r_t = \text{match}_t - \lambda\,\text{cost}_t$ (fidelity vs. compute).

Trained by imitation, then RL. The target q is used only to compute rewards, so the gate is q-free at inference.



Reinforcement-learning formulations

Policy gradient (all variants, with entropy bonus $\eta\,\mathcal{H}[\pi_\theta]$):

$$\nabla_\theta J = \mathbb{E}\Big[\sum_t A_t \, \nabla_\theta \log \pi_\theta(a_t \mid s_t)\Big]$$

differing in the **advantage** A_t:

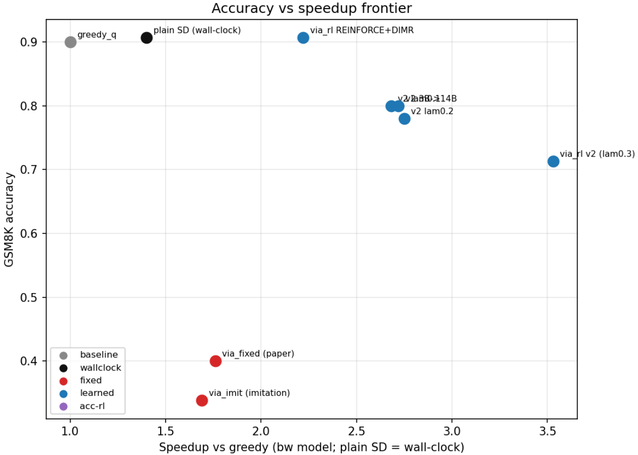
- **REINFORCE**: $A = R - b$ (global EMA baseline).
- **GRPO**: $A_i = (R_i - \mathrm{mean}_{\mathcal{G}} R)/\mathrm{std}_{\mathcal{G}} R$, with PPO clipping and KL to the warm-start.
- **Per-token**: $\hat{A}_t = (r_t - \mathrm{mean}_{\text{batch}}\, r)/\mathrm{std}_{\text{batch}}\, r$.

Results

Method	GSM8K acc	q-calls/tok	speedup
greedy-q (reference)	0.887	1.000	1.00×
plain SD (lossless)	0.907	0.259	1.40×
via_fixed (paper, tuned)	0.40	0.31	1.76×
via_imit (imitation only)	0.34	0.29	1.69×
via_rl REINFORCE	0.880	0.243	2.29×
via_rl GRPO	0.853	**0.162**	2.94×
via_rl REINFORCE+DIMR	**0.907**	0.250	2.22×
via_rl v2 per-token	0.713	0.104	**3.53**×

⇒ **S^{4}D is state-of-the-art hierarchical speculative decoding.**

The accuracy and speed frontier



Conclusion

- S^{4}D replaces VIA-SD's **hand-tuned thresholds** with a **learned per-token policy**.
- **Pareto-dominates** the baseline: matches *lossless* accuracy (0.91) at **4**$\times$ **fewer** full-model calls, up to **3.5**$\times$ speedup.
- Deploys as a small q-free MLP, and **improves with stronger drafters**.

A new state of the art in hierarchical speculative decoding.

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Thank you. Questions?

Backup: ablations

Setup	GSM8K acc	q-calls/tok	note
0.5B $\rightarrow$ 14B (main)	0.88–0.91	0.10–0.25	baseline pair
3B $\rightarrow$ 14B (bigger drafter)	0.80	**0.028**	**36**$\times$ **fewer** calls
0.5B $\rightarrow$ **32B** (bigger verifier)	**0.33**	0.07	accuracy collapses

- A DIMR-searched q' lifts REINFORCE 0.88 $\rightarrow$ 0.91 (moot once the policy is accept-heavy).
- Optimizing final-answer correctness directly did not surpass the match-trained policy.

Backup: future work

- **Spec-spec decoding:** a recursive drafter hierarchy; the gate routes across levels.
- **KnapSpec:** allocate a fixed budget of full-model calls as a knapsack over the highest-value tokens.
- **Correctness-floor RL:** minimize cost subject to an accuracy floor.
- **Scale:** bigger drafters, more tasks, real-engine wall-clock.